ADITYA MITTAL

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WORK EXPERIENCE

Data Science Capstone Consultant
Sept 2026 — Present
Delta Dental of Michigan, Ohio & Indiana — *East Lansing, MI*

- **Self-Service Data Ingestion & Insights Application:** Designing a self-service data ingestion and insights application in Dataiku Cloud enabling non-technical users to upload structured (CSV/Excel) and unstructured (PDF) files and receive automated analysis without writing code; architecting a multi-recipe Dataiku Flow spanning schema inference, data cleaning, and a reusable schema/template registry to eliminate redundant re-detection on known file types
- **Unstructured Extraction, NL Querying & Delivery:** Implementing unstructured document extraction via Dataiku's document-handling and LLM Mesh capabilities to normalize PDF content into the same schema as tabular sources; building a natural-language querying layer with Dataiku Prompt Studio and a guided no-code front end (Dataiku Application Designer, with a Streamlit-based alternative), with Dataiku Scenarios automating ingestion on new file arrival
- **Client Engagement:** Collaborating cross-functionally with Delta Dental's data science team through a semester-long, client-facing engagement, translating open-ended business requirements into a scoped technical roadmap

Research Assistant, Broadband Infrastructure & Policy Analytics
Jun 2026 — Present
Michigan State University, Quello Center (Prof. Johannes Bauer) — *East Lansing, MI*

- **BEAD Program Tracking System:** Architected a database-backed BEAD tracking system (SQLite, automated Excel reporting); designed a 3-tier source-reliability framework with automated cross-validation that caught a real data error and resolved an unconfirmed claim against authoritative federal data
- **Data Coverage & Stakeholder Dissemination:** Engineered a two-tier keyword-matching pipeline for automated news/RSS ingestion, closing a coverage gap for BEAD-relevant reporting lacking explicit program terminology; integrated NTIA's national BEAD approval dataset (Aug 2026), extending verified coverage from 4 to all 56 states and territories, and presented methodology and findings to the faculty PI and Louisiana state government data science staff
- **County-Level Broadband Modeling (R & Python):** Extended a fixed-effects panel regression (Census ACS, FCC data, ~3,100 US counties, 2022–2024) by resolving a structural modeling limitation via pooled ML methods; implemented and benchmarked LASSO, Random Forest, and XGBoost to identify predictors of broadband access, evaluated via R^2 and RMSE
- **Interactive National Broadband Data Explorer (D3.js/TopoJSON):** Designed and built a progressive-disclosure national-to-county drill-down visualization combining a choropleth and proportional-symbol layer with real-time federal funding data; integrated a validated 100K+ row dataset adding a 3-tier broadband coverage classification, catching a technology-inclusion bias methodology error via cross-validation before deployment

Data Scientist
Jul 2024 — Jan 2025
Aavas Financiers Ltd. — *Jaipur, India*

- **Churn Prediction & Retention Modeling:** Developed an XGBoost churn prediction model on a 400K+ customer bureau dataset; applied SMOTE to correct thin-file class imbalance and boosted minority class recall by 11% over the logistic regression baseline via Optuna hyperparameter tuning
- **Operational Performance Dashboard:** Built Python and SQL pipelines automating data ingestion and dashboard delivery across 438 branches, reducing reporting lag from T+2 to same-day; eliminated 3+ hours of daily manual analyst work and owned data quality standards for production reporting
- **Stakeholder Communication:** Translated complex model outputs into actionable insights for senior leadership through structured dashboards and reports; collaborated cross-functionally with business and engineering teams to implement data-driven decisions

Data Analysis Research Intern
Jun 2023 — Sept 2023
MNIT Jaipur — *Jaipur, India*

- **Fraud Detection Pipeline:** Built an end to end ML pipeline on 500K+ transaction records; engineered 10+ behavioral features, applied SMOTE to address 97:3 class imbalance, and optimized XGBoost via Optuna improving F1-score by 15% with full methodology documentation for research team handoff

PROJECTS

FinSight: Financial Document Intelligence System Streamlit
Jan 2026 — Apr 2026

- **RAG Pipeline & LLM Evaluation:** Built an end to end Retrieval-Augmented Generation system (LlamaIndex, ChromaDB, Anthropic API) over 50+ SEC 10-K filings; implemented hybrid dense and BM25 retrieval with 512-token overlapping chunk windows, evaluated 5 pipeline configurations via RAGAS achieving Faithfulness > 0.85 on a 25-question hand-labeled analyst eval set; deployed as a Streamlit dashboard with inline source citations

Credit Risk Prediction Dashboard Streamlit
Sept 2025 — Dec 2025

- **End to End ML Product:** Identified a structural inefficiency in manual loan scoring workflows and replaced it with an automated pipeline: SMOTE class balancing, XGBoost and Logistic Regression modeling (82% AUC), SHAP waterfall explainability, and an interactive Streamlit dashboard with live performance monitoring; deployed and used by non-technical stakeholders

EDUCATION

Master of Science, Data Science
Aug 2025 — May 2027
Michigan State University, GPA: 3.5/4.0 — *East Lansing, MI*

Bachelor of Technology, Computer Science and Engineering
Oct 2020 — May 2024
SRM University KTR, GPA: 3.58/4.0 — *Chennai, India*

SKILLS

Machine Learning: XGBoost, SHAP, SMOTE, CNN, Transfer Learning, Feature Engineering, Hypothesis Testing, Statistical Modeling, Regression, Clustering, Anomaly Detection, scikit-learn, TensorFlow/Keras, PyTorch (learning)
LLM & Agentic AI: Anthropic API, LlamaIndex, RAG Pipelines, Hybrid Retrieval (Dense + BM25), ChromaDB, RAGAS Evaluation, Prompt Engineering, Multi-Agent Architecture, Vector Embeddings, FAISS, Dataiku LLM Mesh, Claude Code, Cursor
Data & Engineering: Python, SQL, R, Git/GitHub, Pandas, NumPy, Pipeline Automation, Dataiku, GeoDa, ArcGIS, Streamlit, Tableau
Analytics & Communication: A/B Testing, Cohort Analysis, KPI Design, EDA, Stakeholder Communication